

TG-MPPI control cycle

environment-derived ancillary controllers + mode-conditioned path-integral control

Inputs

robot state x_k

local costmap C_k

global path $\mathcal{P}$

tracked obstacles $\hat{O}_{0:T}$

1 Topology proposals

Dijkstra geodesic field d_r

body B_p + membrane

pseudopod backtraces P_m

space--time wait / detour

winding distinctness filter

constraint projection + footprint gate

outputs: $\mu_0, \dots, \mu_M$

$\{\mu_m\}$

2 Grouped MPPI

allocate N_m samples per mode

$U^{(i)} \sim \mathcal{N}(\mu_m, \Sigma_m)$

parallel dynamics rollout

Nav2 static/path critics

predicted-obstacle critic

conditional weights $w_{i|m}$

free energy F_m

$\{F_m, w_{i|m}\}$

3 Decision \& execution

$m^* = \arg \min_m F_m$

dwelling + switch margin

multi-cycle confirmation

selected-group weighted update

apply first command u_k^*

shift warm start

repeat at next control cycle

u_k^*

receding-horizon feedback